MODULE 29 · DOMAIN 5 - IMPLEMENT INFORMATION EXTRACTION SOLUTIONS

Ingesting and indexing for grounding

Data sources and indexers, change and deletion detection, document cracking, images, audio and video, permissions, and knowledge sources.

DOMAIN 5 - Information extraction WEIGHT 10-15% MODULE 29 of 39

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Grounding starts **before** the query. Whatever the indexer didn't ingest, retrieval can never find.

From source system to search index

01
Data source
Where the content lives.

02
Crack
Extract text, images, metadata.

03
Enrich
Skills: OCR, chunking, vectors.

04
Map
Fields into the index schema.

05
Schedule
Run on demand or on a timer.

06
Retrieve
Queries and agents, later.

ON THE EXAM

An indexer is the engine for the middle four steps. Module 30 covers enrichment skills in depth.

Supported data sources

SOURCE	NOTES
Azure Blob Storage	The most common source; automatic change detection
Azure Data Lake Storage Gen2	Hierarchical folders; can carry ACL permissions
Azure SQL and Cosmos DB	Rows become documents; needs a change detection policy
Azure Table Storage and Files	Rows and file shares
Microsoft OneLake, SharePoint	Lakehouse files and sites
Anything else	Logic Apps connectors, or push documents yourself

ON THE EXAM

Indexers pull on a schedule. The push model sends documents to the index directly, for near-instant updates.

How an indexer runs

- An indexer joins a **data source**, an **index**, and optionally a **skillset**.
- Run it **on demand**, or on a schedule as often as **every five minutes**.
- **Field mappings** connect source fields to index fields when names or types differ.
- Progress is tracked with an internal **high water mark**, so later runs process only changes.
- A service runs **one indexer job per search unit**, and heavy indexing can slow queries.

> ON THE EXAM
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> Content that must appear within seconds can't wait for a five-minute schedule. That's a push-model requirement.

Change and deletion detection

SITUATION	WHAT HAPPENS
New or changed blobs	Detected automatically from timestamps
New or changed database rows	Needs a change detection policy in the data source
Deleted source content	Needs a deletion detection policy: it isn't automatic
Native blob soft delete	Blob containers only; not ADLS Gen2 or Azure Files
Renamed ADLS Gen2 directory	Timestamps don't change, so blobs aren't reindexed

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Native soft delete also needs the document key mapped to a blob property, and blob versioning switched off.

Deletion detection isn't automatic, and it isn't **retrospective**. Set the policy **before the first run**.

What comes out of a file

- Indexers extract **text, images and metadata** from files like PDFs and Office documents.
- **Scanned pages have no text layer.** Without OCR, they arrive empty.
- Set **imageAction** to extract embedded images for processing.
- Up to **1,000 images** per document are extracted.
- OCR and image analysis are free for **20 transactions per indexer per day**, then billed.

> THE SIGNAL
>
> 'Our PDFs return nothing' usually means scanned pages with no OCR in the skillset.

Search doesn't look at images at **query** time. Whatever you want found must become **text or vectors** at indexing time.

Three ways to make images searchable

APPROACH	WHAT IT PRODUCES	SUITS
OCR and image analysis	Text found in the image, tags, captions	Scans, screenshots, text in pictures
Verbalization (GenAI Prompt skill)	An LLM's description of the image	Diagrams, charts, workflow images
Multimodal embeddings	A vector for the image itself	Visual similarity, photo search

ON THE EXAM

Store the image's location too, so an answer can show the diagram it cites.

Ingesting recordings

- Indexers don't transcribe media. Convert it to text **before** indexing.
- **prebuilt-audioSearch**: transcription with **speaker diarization**, timings and summaries.
- **prebuilt-videoSearch**: frames, transcription, summaries, and **segments** aligned in time.
- Index each **segment** as a document, keeping its **timestamps** for citations.
- Azure Speech batch transcription is an alternative for plain audio, as module 28 showed.

> ON THE EXAM
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> Segment-level documents let an answer cite 'from 12:04 to 15:30 of the onboarding video'.

Carrying access rights into the index

- **ADLS Gen2 ACLs** and **blob RBAC scopes** can be ingested as **permission metadata**.
- **Knowledge sources** carry document-level permissions forward into indexed content.
- At query time, results are then filtered by the signed-in user's access, as module 13 showed.
- Without permission metadata, one index means **everyone sees everything** in it.
- The alternative is separate indexes per audience, which multiplies maintenance.

> ON THE EXAM
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> Permission-aware ingestion is what makes a single company-wide index safe to query.

The managed path to a grounded index

- Creating a knowledge source generates a **data source, skillset, index and indexer** for you.
- The skillset **chunks** and optionally **vectorizes** multimodal content.
- Types include **blob** (Blob and ADLS Gen2), OneLake, SharePoint, and an existing search index.
- Blob, OneLake and SharePoint sources can use **Content Understanding** for content extraction.
- Knowledge sources feed **knowledge bases**, the Foundry IQ layer from module 03.

> ON THE EXAM
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> Agentic retrieval is generally available in the 2026-04-01 REST API; the portals remain preview.

Knowledge source or hand-built pipeline?

KNOWLEDGE SOURCE OR WIZARD

Managed defaults

- Objects created for you, in minutes
- Chunking, vectorisation and permissions included
- Less to maintain, fewer choices

HAND-BUILT PIPELINE

Full control

- Choose every skill, field and mapping
- Custom chunking, custom skills, custom fields
- More to build and keep working

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Start managed, and drop to a hand-built pipeline only where a requirement demands it.

Ingestion symptoms and causes

SYMPTOM	LIKELY CAUSE
New documents never appear	Indexer not scheduled, or not run
Deleted documents still answer queries	No deletion detection policy
Scanned PDFs return no text	No OCR or Content Understanding in the skillset
Diagrams never match a query	Images not verbalized, vectorized or OCR'd
Videos can't be cited by moment	Segments not indexed with timestamps
Everyone sees restricted files	No permission metadata ingested

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Every row is an ingestion problem. None of them can be fixed by changing the query.

> A support portal indexes 40,000 PDFs from Blob Storage. Staff delete outdated PDFs every week, but deleted documents keep appearing in search results.
>
> The indexer runs nightly, and new and updated PDFs are indexed correctly.

What should the team do?

A Run the indexer more often, so deletions are noticed sooner.

B Reset the existing indexer, which clears its high water mark.

C Switch to the push model and delete documents from the index directly.

D Add a deletion policy, on a new index built with it.

◆ PAUSE AND DECIDE

What should the team do?

A Run more often.

More runs of a pipeline that never detects deletions at all.

B Reset the indexer.

Microsoft: documents deleted before the policy existed stay, even after a reset.

C Push model.

Rebuilds the whole ingestion approach to solve one missing policy.

D Deletion policy, new index from the start.

Deletion detection isn't automatic or retrospective, so start clean with the policy.

THE SIGNAL

New and updated documents fine, deletions ignored: the missing piece is a deletion detection policy.

> An engineering wiki's PDFs contain architecture diagrams with very little surrounding text. Searches for concepts shown only in the diagrams return nothing.
>
> The team wants answers to describe the relevant diagram, and show it to the reader.

What should the team add to ingestion?

A Verbalize each image with the GenAI Prompt skill; store locations.

B Increase the chunk overlap so diagram captions appear in more chunks.

C Switch retrieval to a semantic ranker so diagram pages rank higher.

D Store the PDFs in a second container and index them a second time.

◆ PAUSE AND DECIDE

What should the team add to ingestion?

A Verbalize images; store locations.

Descriptions become searchable text, and the stored location lets answers show the diagram.

B More chunk overlap.

Overlaps text that barely exists. The content is in the picture.

C Semantic ranker.

Reranks what retrieval found. Nothing represents the diagrams yet.

D Index twice.

Two copies of the same unreadable diagrams.

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Search never reads images at query time. Diagram meaning has to become text or vectors during ingestion.

> A training team has 900 recorded webinars. Staff should be able to search by topic and jump to the right moment in a recording.
>
> The recordings are MP4 files in Blob Storage, and the current indexer produces only file names and metadata.

What should the team do?

A. Point the indexer directly at the MP4 files and enable OCR on them.

B. Run an image analysis skill over every frame of every recording.

C. Analyse with Content Understanding; index timestamped segments.

D. Ask presenters to upload their slide decks, and index those instead.

◆ PAUSE AND DECIDE

What should the team do?

A OCR the MP4s.
Indexers don't transcribe media, and OCR reads images, not audio.

B Analyse every frame.
Enormous cost, no speech, and no segment structure.

C Content Understanding, then index segments.
Transcripts, summaries and time-aligned segments make moments searchable.

D Index the slides.
Loses what was said, and still can't jump to a moment in the video.

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Media has to become text before indexing. Segments with timestamps are what make 'jump to the moment' possible.

> A company wants one grounded knowledge base over departmental files in ADLS Gen2, for an agent used by all staff. Folder permissions already restrict who can read what.
>
> The team is small and wants the least plumbing to build and maintain.

What should the team use?

A Separate indexes per department, each with its own indexer and schedule.

B A blob knowledge source, which carries document permissions into the index.

C One index for everything, with a system message forbidding restricted answers.

D A nightly export of permitted files per user into per-user containers.

◆ PAUSE AND DECIDE

What should the team use?

A. Index per department.
Works, but multiplies indexers, schedules and cost for a small team.

B. Blob knowledge source.
Creates the pipeline for you and carries ACL permissions into the index.

C. One index, prompt rule.
Module 17's lesson: instructions aren't access control.

D. Per-user exports.
Copies files per person, and goes stale between nightly runs.

THE TIEBREAKER

Existing folder permissions plus minimal plumbing: a knowledge source with permission metadata.

Ingestion in five lines

- **Retrieval can only find what ingestion put there.**
- **Change detection is automatic for storage; deletion detection isn't.** Set it up front.
- **Images:** OCR, verbalize or embed at indexing time. Search never reads them at query time.
- **Audio and video:** Content Understanding first, then index segments with timestamps.
- **Knowledge sources** build the pipeline and carry permissions into the index.

> ON THE EXAM
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> Next, module 30: enrichment with built-in and custom skills, layout, and integrated vectorisation.